Generating Chest X-rays with a Convolutional Variational Autoencoder

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Abstract

We implement a convolutional variational autoencoder (VAE) in PyTorch and train it on the pneumonia subset of the Kaggle chest X-ray dataset. We report the hyperparameters, the training loss curve, Inception Score (IS) and Fréchet Inception Distance (FID) of the generated images computed with PyTorch-Ignite, and qualitative samples. The model reaches an FID of – and an IS of – against the test set.

1 Method

Model. A VAE [1] learns an encoder $q_\phi(z \mid x)$ and a decoder $p_\theta(x \mid z)$ with a standard-normal prior on a latent $z \in \mathbb{R}^{128}$. The encoder is a four-layer strided convolutional network (4×4 kernels, stride 2, padding 1) with channel widths 32–64–128–256, BatchNorm after every convolution except the first, and LeakyReLU(0.2) activations. It maps a $1\times64\times64$ image to a $256\times4\times4$ feature map, which two linear heads turn into μ and $\log\sigma^2$. Sampling uses the reparameterisation $z = \mu + \sigma \odot \epsilon$, $\epsilon \sim \mathcal{N}(0, I)$. The decoder mirrors the encoder: a linear layer to $256\times4\times4$, then four transposed convolutions (256–128–64–32–1) with BatchNorm and ReLU, ending in a sigmoid so outputs lie in $[0, 1]$. The full model has 2.96M parameters (1.74M encoder, 1.22M decoder).

Objective. We minimise the negative evidence lower bound, summed over pixels and the batch:

$$\mathcal{L} = \underbrace{\sum_i \mathrm{BCE}(\hat{x}_i, x_i)}_{\text{reconstruction}} + \underbrace{-\tfrac{1}{2}\sum_j \left(1 + \log\sigma_j^2 - \mu_j^2 - \sigma_j^2\right)}_{\text{KL to } \mathcal{N}(0,I)}. \tag{1}$$

BCE is appropriate since inputs are in $[0, 1]$ and the decoder is sigmoid-bounded.

Data. We use the chest X-ray pneumonia dataset [2]. The repository contains the PNEUMONIA class only: 3875 training and 390 test images. Each image is converted to a single grey channel, resized to 64×64 for compute reasons, and scaled to $[0, 1]$. No augmentation is used. All code (`VAE/`, `train.py`, `eval_model.py`, `data.py`, `main.py`, `train_gpu.sbatch`) is submitted with this report.

2 Hyperparameters

Table 1 lists every setting; training ran on one RTX 2080 Ti GPU.

3 Results

Training loss. Figure 1 shows the per-image training loss (reconstruction plus KL) over 100 epochs. The loss falls from – after the first epoch to – at the end, with a minimum of – at epoch –.

Setting	Value
Input resolution	64×64, 1 channel
Latent dimension	128
Optimiser	Adam, learning rate 10^{-3}, default β
Batch size	32
Epochs	100
Loss	BCE (sum) + KL, equal weight ($\beta = 1$)
Normalisation / activations	BatchNorm; LeakyReLU(0.2) encoder, ReLU decoder
Data loading	8 workers, shuffled every epoch

Table 1: Hyperparameter settings.

Inception Score and FID. Following the PyTorch-Ignite GAN-evaluation recipe, we draw $z \sim \mathcal{N}(0, I)$, decode one fake image for every real test image, and update Ignite's `FID` and `InceptionScore` metrics batch by batch. Both metrics use the ImageNet-pretrained Inception-v3 network, so the single grey channel is replicated to three channels and the 64×64 images are bilinearly upsampled to 299×299 before feature extraction. Table 2 reports the scores. The IS is near its lower bound of 1, as expected: ImageNet Inception assigns nearly the same class distribution to every chest X-ray, real or generated. FID, which compares feature statistics of generated and real images directly, is the more informative number here.

Metric	Value	Better
FID (vs. 390 test images)	–	lower
Inception Score	–	higher

Table 2: Evaluation of 390 generated samples with PyTorch-Ignite.

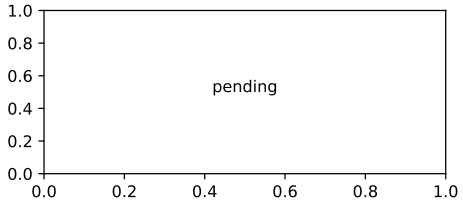


Figure 1: Training loss per image per epoch.

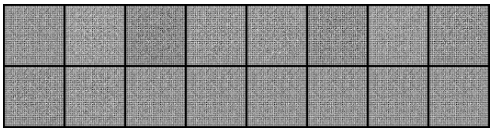


Figure 2: Sixteen samples decoded from $z \sim \mathcal{N}(0, I)$.

Generated images. Figure 2 shows sixteen random samples. The model captures the global anatomy of a frontal chest radiograph: the dark lung fields, the bright mediastinum and spine, and the shoulder and rib outline. As is typical of a VAE trained with a pixel-wise likelihood, the samples are blurry and lack fine detail such as individual ribs or vessel markings. Higher resolution, a perceptual reconstruction term, or a tempered KL weight are natural next steps.

References

[1] D. P. Kingma and M. Welling. Auto-encoding variational Bayes. *ICLR*, 2014.

[2] D. S. Kermany et al. Identifying medical diagnoses and treatable diseases by image-based deep learning. *Cell*, 2018. `https://www.kaggle.com/paultimothymooney/chest-xray-pneumonia`.

[3] PyTorch-Ignite. GAN evaluation with FID and IS. `https://pytorch-ignite.ai/blog/gan-evaluation-with-fid-and-is/`.